Cote 140-2

[scan 341]

Of course, this character will again be called, "multiplier", and|or "cyclotomic character". † If our fundamental conjecture is true, this character identifies $(\Gamma_\Sigma)_{\mathrm{ab}}$ with $\widehat{\mathbb{Z}}^*$.

On the other hand, if $I = I(\pi)$, the canonical homomorphism

$$\widehat{\widehat{\mathcal{T}}} \longrightarrow \mathfrak{S}_I | \mathcal{S}_I$$

is † not trivial on Σ, but induces a surjective homomorphism

$$\Sigma \longrightarrow \mathfrak{S}_I^{\dagger}$$

whence a subgroup $\Sigma^!$ such that

$$\Sigma/\Sigma^! \xrightarrow{\sim} \mathfrak{S}_I.$$

We see at once that $\forall\ \gamma|g \in \widehat{\widehat{\mathcal{T}}}$ which normalizes Σ normalizes $\Sigma^! = \Sigma \cap \widehat{\widehat{\mathcal{T}}}^!$, whence|i.e. $\Sigma^!$ is also invariant in $\mathcal{N}_\Sigma$. † Let $\mathcal{N}_\Sigma^! = \mathcal{N}_\Sigma \cap \widehat{\widehat{\mathcal{T}}}^!$, we have a Cartesian diagram of subgroups of $\widehat{\widehat{\mathcal{T}}}$

morally, $\Sigma^! \simeq \widehat{\mathcal{T}}^{!+}$

$$\begin{array}{ccc} \mathcal{N}^! & \xhookrightarrow{\mathfrak{S}_I} & \mathcal{N} \\ {\scriptstyle \Gamma_\Sigma}\uparrow & & \uparrow{\scriptstyle \Gamma_\Sigma} \\ \Sigma^! & \xhookrightarrow[\mathfrak{S}_I]{} & \Sigma \end{array}$$

giving an injective homomorphism

$$\mathcal{N}/\Sigma^! \longrightarrow \underbrace{\mathcal{N}/\mathcal{N}^!}_{\mathfrak{S}_I} \times \underbrace{\mathcal{N}/\Sigma}_{\Gamma_\Sigma}$$

of which|when we see at once that it is bijective [m2]

Marginal note. [m2] and|for $\Sigma/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\mathcal{N}^!$ $(\simeq \mathfrak{S}_I)$

$$\mathcal{N}/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\mathcal{N}^! \times \mathcal{N}/\Sigma \simeq \mathfrak{S}_I \times \Gamma_\Sigma$$

and consequently we also have

$$\mathcal{N}^!/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\Sigma = \Gamma_\Sigma,$$

i.e. the group Γ_Σ of arithmetically outer automorphisms of (π, Σ) can also be described via the automorphisms [p. 340]

Uncertainty retained from cancelled passages.

140-2-s341-D001 It

140-2-s341-D002 [illeg. 1]

140-2-s341-D003 !

140-2-s341-D004 [illeg. 3]

[scan 342]

that are outer and induce the identity on I. [p. 173]

In the case $(g, \nu) = (0, 3)$, we have $\Sigma^! = (1)$, ⇒ hence $\Sigma \xrightarrow{\sim} \mathfrak{S}_3$. $\mathcal{N}$ is the normalizer of $\mathfrak{S}_3$ in $\widehat{\widehat{\mathcal{T}}}$, $\mathcal{N}^! \xrightarrow{\sim} \Gamma_\Sigma$ is its centralizer, and $\mathcal{N}$ is identified with the direct product of the two.

†

Now let $I|I' \subset I$, $\nu' = \operatorname{card} I'$, and let us consider the outer group π' obtained from π by "plugging the holes at $I \setminus I'$", whence a homomorphism

$$\pi^* \longrightarrow \pi'.$$

† Let us consider the bases of π, for which † the ν' first l_i, are in † groups with loops † $L_{i'}$ $(i' \in I')$ — we call them adapted to I'; they form a torsor $\subset \operatorname{Isom}_{\text{lac}}(\hat{\pi}_{g,\nu}, \pi)$ under the subgroup of † $\widehat{\widehat{\mathcal{G}}}_{g;(\nu,\nu'\dagger)}$ of $\widehat{\widehat{\mathcal{G}}}_{g,\nu} = \operatorname{Aut}_{\text{lac}}(\hat{\pi}_{g,\nu})$ consisting of the $u|h \in \widehat{\widehat{\mathcal{G}}}_{g,\nu}$ whose image in $\mathfrak{S}_I$ leaves invariant † the set of the ν'† first elements (or, of the $\nu - \nu'$ last its|their complementary set,).

m3

Marginal note. m3 Whence also: ∀ basis, ∀ discretification, oriented discretification of π defines one of π', whence ditto for classes of π, π' conjugation, i.e. for|by the strict prediscretifications (possibly oriented), and also for|by the closures of the classes, i.e. the † prediscretifications and oriented prediscretifications. It is only the case of arithmetizations|arithmetizations that † would require a finer analysis…

For such a basis, we find a corresponding basis of π'. The datum of (π, I') is equivalent to that of a torsor under $\widehat{\widehat{\mathcal{G}}}_{g,(\nu,\nu'\dagger)}$, that of a π' to † the datum of a torsor under $\widehat{\widehat{\mathcal{G}}}_{g,\nu'}$, and the passage from π to π' is described by the change of group of operators

$$\widehat{\widehat{\mathcal{G}}}_{g,(\nu,\nu')} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu'}$$

this homomorphism sends $\widehat{\mathcal{G}}_{g,(\nu,\nu')} = \Sigma_{g,(\nu,\nu')}$ into $\widehat{\mathcal{G}}_{g,\nu'} = \Sigma_{g,\nu'}$, [p. 341]

$\Sigma_{g,\nu}^{\cap}$

Uncertainty retained from cancelled passages.

140-2-s342-D002 [illeg. 2]
140-2-s342-D003 For a map
140-2-s342-D005 [illeg. 2]
140-2-s342-D007 bases
140-2-s342-D008 in
140-2-s342-D009 [illeg. 1]
140-2-s342-D010 +2
140-2-s342-D012 also
140-2-s342-D013 +2
140-2-s342-D014 [illeg. 1]
140-2-s342-D015 requires
140-2-s342-D017 +2
140-2-s342-D018 [illeg. 1]

[scan 343]

and even $\mathcal{G}_{g,(\nu,\nu')}$ into $\mathcal{G}_{g,\nu'}$, also π into π'. It follows that every prediscretification of π defines one of π', the same for strict prediscretifications, discretifications, outer bases "adapted to I'" †. Finally, if we have a "prearithmetization" of π, i.e. a $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, do we deduce from it a prearithmetization of π' — i.e. (for example) if two prediscretifications of π are compatible i.e. † have the same group $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, of outer automorphisms, is the same true of their images?

†

To see more clearly, we are going to write in diagram form the remarkable sets associated with π, and this in two ways, one without using any particular structure|particular structures on $I = I(\pi)$, the other using a total order, which makes it possible, in the torsor $\operatorname{Isom}_{\text{lac}}(\hat{\pi}_{g,\nu}, \pi)$ under $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, to define the subtorsor under $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$ which we may denote † $\operatorname{Isom}^{!}_{\text{lac}}(\hat{\pi}_{g,\nu}, \pi)$. [p. 342]

Uncertainty retained from cancelled passages.

140-2-s343-D002 (i.e.) for which

140-2-s343-D003 define

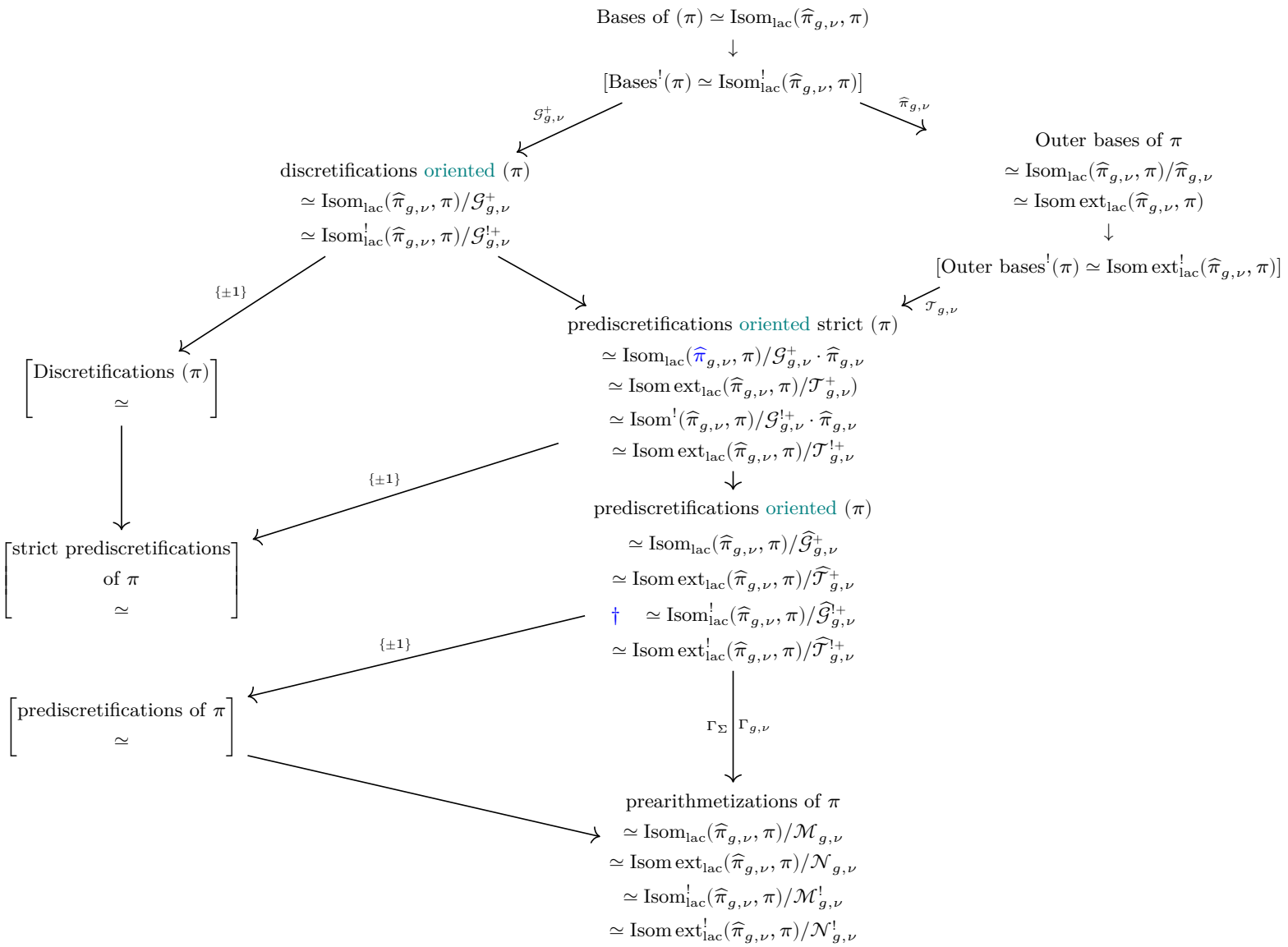
140-2-s343-D004 [ ] [which sends Σ

140-2-s343-D005 [illeg. 1]

[scan 344]

[Diagram continued on the landscape page; 140-2-344-D1]

[p. 174]


Bases of $(\pi) \simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Bases}^{!}(\pi) \simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\mathcal{G}^{+}_{g,\nu}$
$\widehat{\pi}_{g,\nu}$
discretifications oriented (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu}$
Outer bases of π
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Outer\ bases}^{!}(\pi) \simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\{\pm 1\}$
$\mathcal{T}_{g,\nu}$
prediscretifications oriented strict (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{+}_{g,\nu})$
$\simeq \mathrm{Isom}^{!}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{!+}_{g,\nu}$
Discretifications (π)
$\simeq$
$\{\pm 1\}$
prediscretifications oriented (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{+}_{g,\nu}$
$\dagger \quad \simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{!+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{!+}_{g,\nu}$
strict prediscretifications of π
$\simeq$
$\{\pm 1\}$
Γ_{Σ}
$\Gamma_{g,\nu}$
prediscretifications of π
$\simeq$
prearithmetizations of π
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}^{!}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}^{!}_{g,\nu}$

m7

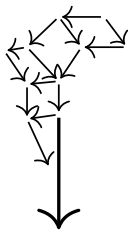
not separated
not separated

Marginal note. m7



m8

Marginal note. m8



Please turn over
[p. 343]

Uncertainty retained from cancelled passages.
140-2-s344-D002 ≃

[scan 345]

N.B.
We set †

$$\Gamma_{g,\nu} = \mathcal{N}_{g,\nu}/\Sigma_{g,\nu} = \mathcal{N}^{!}_{g,\nu}/\Sigma^{!}_{g,\nu}$$

(where $\Sigma_{g,\nu} = \widehat{\mathcal{T}}^{+}_{g,\nu}$, $\Sigma^{!}_{g,\nu} = \widehat{\mathcal{T}}^{!+}_{g,\nu}$). We can also describe it as the group Γ_Σ associated with a profinite outer group with loops π equipped with an oriented prediscretification α (nothing more). If we have two such pairs $(\pi, \alpha), (\pi', \alpha')$, whence Γ_Σ and $\Gamma_{\Sigma'}$ ($\Sigma = \operatorname{Aut}\operatorname{ext}_{\mathrm{lac}}(\pi, \alpha)$, $\Sigma' = \operatorname{Aut}\operatorname{ext}_{\mathrm{lac}}(\pi', \alpha')$), we indeed find a canonical isomorphism

$$\Gamma_\Sigma \xrightarrow{\sim} \Gamma_{\Sigma'}$$

by taking any outer isomorphism with loops u of π with π', transforming α into α' — the associated isomorphism $\Gamma_\Sigma \longrightarrow \Gamma_{\Sigma'}$ obviously does not depend on the choice of u.

This being said, the pairs (π, Σ) of a profinite|profinite group|groups with loops of type (g, ν), equipped with a prearithmetization, with as morphisms the arithmetically outer isomorphisms (relative to Σ and Σ' ...), form a connected groupoid $\Pi_{g,\nu}$, having an "origin" object (defined up to unique isomorphism as being $(\hat{\pi}_{g,\nu}, \Sigma_{g,\nu})$, or at will any (π, Σ_α) coming from a (π, α) (α an oriented prediscretification of π), and whose group of automorphisms is precisely $\Gamma_{g,\nu}$. [p. 344]

Uncertainty retained from cancelled passages.
140-2-s345-D001 [illeg. 1]

[scan 346]
[p. 175]

We observe that, with the exception of the two homogeneous spaces (under $\widehat{\widehat{\mathcal{G}}}(\pi)$, but not necessarily|necessarily under $\widehat{\widehat{\mathcal{G}}}(\pi, I')$, or even under $\widehat{\widehat{\mathcal{G}}}^{!}(\pi)$) Bases (π) and Outer bases (π), the other four are in fact homogeneous spaces under $\widehat{\widehat{\mathcal{G}}}^{!}$, and are expressed as quotients of the $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$-torsor $\mathrm{Isom}^{!}(\hat{\pi}_{g,\nu}, \pi)$, by the four subgroups

$$\mathcal{G}^{!}_{g,\nu}, \quad \mathcal{G}^{!}_{g,\nu} \cdot \hat{\pi}_{g,\nu}, \quad \widehat{\mathcal{G}}^{!}_{g,\nu}, \quad \mathcal{M}^{!}_{g,\nu}$$

(the last defined as the inverse image of $\mathcal{N}^{!}_{g,\nu}$ in $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, just as $\mathcal{M}_{g,\nu}$ is defined as the inverse image of $\mathcal{N}_{g,\nu}$). We find on the other hand, if|for $I' = \{i_0, \dots, i_{\nu'-1}\}$, a|some homomorphism

$$\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu'},$$

sending $\mathcal{G}^{!}_{g,\nu}$ into $\mathcal{G}^{!}_{g,\nu'}$, † by sending $\pi_{g,\nu}$ into $\pi_{g,\nu'}$, thus $\hat{\pi}_{g,\nu}$ into $\hat{\pi}_{g,\nu'}$, $\widehat{\mathcal{G}}^{!}_{g,\nu} = \Sigma^{!}_{g,\nu}$ into $\widehat{\mathcal{G}}^{!}_{g,\nu'} = \Sigma^{!}_{g,\nu'}$, and the subgroup $\mathcal{G}^{!}_{g,\nu} \cdot \hat{\pi}_{g,\nu}$ of † $\widehat{\mathcal{G}}^{!}_{g,\nu}$ into the subgroup $\mathcal{G}^{!}_{g,\nu'} \cdot \hat{\pi}_{g,\nu'}$ of $\widehat{\mathcal{G}}^{!}_{g,\nu'}$. Finally, since $\Sigma^{!}_{g,\nu}$ maps into $\Sigma^{!}_{g,\nu'}$, † $\mathcal{N}^{!}_{g,\nu}$ maps into the normalizer of $\Sigma^{!}_{g,\nu'}$ (in $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu'}$), I say that this is none other than $\mathcal{N}^{!}_{g,\nu'}$. Changing notation, this amounts to the

Lemma The $\mathcal{N}^{!}_{\Sigma} = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}^{!}}(\Sigma^{!})$ † $(= \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!}) \cap \widehat{\widehat{\mathcal{T}}}^{!})$. not proved †

Obviously we have $\mathcal{N}^{!}_{\Sigma} \subset \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!}) \cap \widehat{\widehat{\mathcal{T}}}^{!}$, conversely let $g \in \widehat{\widehat{\mathcal{T}}}^{!}$ which normalizes $\Sigma^{!}$, let us show that it normalizes Σ i.e. that it is $\in \mathcal{N}$ (thus $\in \mathcal{N}^{!} = \mathcal{N} \cap \widehat{\widehat{\mathcal{T}}}^{!}$). [p. 345]

Uncertainty retained from cancelled passages.

140-2-s346-D001 [illeg. 1] $\widehat{\mathcal{G}}$ $\widehat{\pi}_{g,\nu}$

140-2-s346-D002 [illeg. 1]

140-2-s346-D003 [illeg. 1]

140-2-s346-D005 [illeg. 1]

140-2-s346-D006 [illeg. 4] $\mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!})/\Sigma^{!}$

[scan 347]

It's not clear. I nevertheless feel like proving|pronouncing † the conjecture The canonical map

$$\text{Prediscretifications|Prediscretifications}^{+}(\pi) \xrightarrow{\sim} \text{Prediscretifications|Prediscretifications}^{+}(\pi')$$

is bijective. In order that two oriented prediscretifications α, β of π † have the same group of outer automorphisms $\Sigma_{\alpha} = \Sigma_{\beta}$, it is necessary and sufficient † that it be so. for their images $\Sigma_{\alpha'}$ and $\Sigma_{\beta'}$, i.e. that $\Sigma_{\alpha'} = \Sigma_{\beta'}$; so that the preceding bijection induces a bijection

$$\mathrm{Prearithmetizations}(\pi) \xrightarrow{\sim} \mathrm{Prearithmetizations}(\pi').$$

Finally, the preceding bijections are compatible with the homomorphism of the groups of operators $\widehat{\widehat{\mathcal{T}}}(\pi, I') \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi')$, † a fortiori a $\widehat{\widehat{\mathcal{T}}}^{!}(\pi) \longrightarrow \widehat{\widehat{\mathcal{T}}}^{!}(\pi')$, which implies that it thereby induces|that it induces them (for a given prearithmetization $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$ of π, giving $\Sigma' \subset \widehat{\widehat{\mathcal{T}}}(\pi')$ in π'; with $\Sigma^{!} \longrightarrow \Sigma'^{!}$) a homomorphism $\mathcal{N}^{!}_{\Sigma}$† $\longrightarrow \mathcal{N}^{!}_{\Sigma'}$†, † and the corresponding homomorphism

$$\Gamma_{\Sigma} = \mathcal{N}^{!}_{\Sigma}/\Sigma^{!} \longrightarrow \text{†}\Gamma_{\Sigma'} = \mathcal{N}^{!}_{\Sigma'}/\Sigma'^{!}$$

is an isomorphism.

To convince oneself of this it suffices to look at † (choosing $\pi = \pi_{g,\nu}|\widehat{\pi}_{g,\nu}$, I' † reduced to the|being a last element|element of I_ν) the case where $I' = \{i'|i\}$, We then have a homomorphism

$$\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi').$$

(the outer loop automorphisms of π fixing i define well-determined automorphisms of π' — not just outer ones) [p. 346]

Uncertainty retained from cancelled passages.

140-2-s347-D001 [illeg. 1]
140-2-s347-D002 be
140-2-s347-D004 for their images
140-2-s347-D006 which induces
140-2-s347-D008 (Σ)
140-2-s347-D009 (Σ')
140-2-s347-D010 which is likewise
140-2-s347-D012 $\mathcal{N}^{!}_{\Sigma'}/\Sigma'^{!}$
140-2-s347-D013 $\pi = \widehat{\pi}_{g,\nu}$
140-2-s347-D014 [illeg. 1]

[scan 348]

i.e. $\widehat{\widehat{\mathcal{G}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi')$ is trivial on $\widehat{\pi}$, and thus passes to the quotient as a $\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi, i)$. [p. 176] I assume, by analogy with the discrete case, that this homomorphism is an isomorphism, so that we have an exact sequence m11

Marginal note. m11



m12

Marginal note. m12 recall that we assume π anabelian|nonabelian $(2g + \nu \geq 3)$

$$1 \longrightarrow \pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi', i) \longrightarrow 1$$

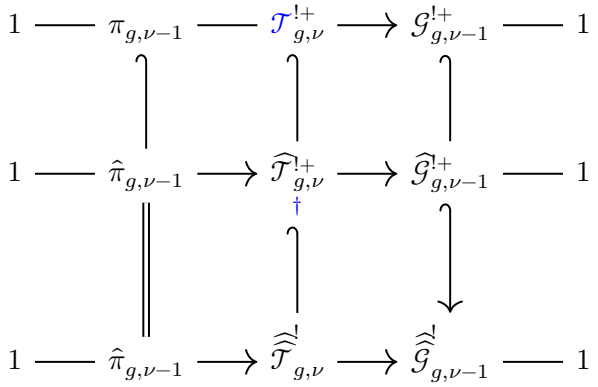
or again

$$1 \longrightarrow \widehat{\pi}_{g,\nu-1} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1} \longrightarrow 1$$

which contains the exact sequence

$$1 \longrightarrow \pi^{*|+}_{g,\nu-1} \longrightarrow \mathcal{T}_{g,(\nu,\nu-1)} \longrightarrow \mathcal{G}_{g,\nu-1} \longrightarrow 1$$

as|known sub†, sequence|is exact. It is convenient to work instead with



The map

[↪ 140-2-348-M1]

Prediscretifications|Prediscretifications$^{+}(\pi)$

$\longrightarrow$ Prediscretifications|Prediscretifications$^{+}(\pi')$

is thus identified with $\widehat{\widehat{\mathcal{F}}}^{!}_{g,\nu}/\widehat{\mathcal{F}}^{!}_{g,\nu}$, we read off the diagram that this is $\simeq \widehat{\widehat{\mathcal{G}}}^{!\dagger}_{g,\nu-1}/\widehat{\mathcal{G}}^{!+}_{g,\nu-1}$, whence bijectivity on the sets of oriented prediscretifications.

In the group $\widehat{\widehat{\mathcal{F}}}^{!}_{g,\nu} = \mathcal{F}$, we have an invariant subgroup $\hat{\pi}_{g,\nu-1} = \pi'$, and a subgroup Σ $(= | \leftarrow \widehat{\mathcal{F}}^{!+}_{g,\nu})$ [p. 347]

Uncertainty retained from cancelled passages.

140-2-s348-D002 particular|dense part

140-2-s348-D003 [*illeg.* 1]

140-2-s348-D004 +

[scan 349]

between π' and $\mathcal{F}$, † giving a subgroup $\Sigma' \simeq \mathcal{F}(\pi')(= \widehat{\mathcal{G}}^{!+}_{g,\nu-1})$ in $\mathcal{F}' = \mathcal{F}/\pi'$ $(\simeq \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu-1})$, and we know well that then

$$\mathcal{F}/\Sigma \simeq \mathcal{F}'/\Sigma',$$

and † that the normalizer $\mathcal{N}$, of Σ in $\mathcal{F}$ is the inverse image of the normalizer $\mathcal{N}'$ of Σ' in $\mathcal{F}'$, so that $\mathcal{N}/\Sigma \simeq \mathcal{N}'/\Sigma'$, which proves what we wanted.

Corollary We have canonical isomorphisms

$$\Gamma_{g,\nu} \overset{\sim}{\to} \Gamma_{g,\nu-1} \overset{\sim}{\to} \cdots \quad \dagger$$

m13

Marginal note. m13 We also see that $\Sigma|\Sigma^{!}$ invariant in $\mathcal{F}|\widehat{\widehat{\mathcal{F}}}$ is equivalent to Σ' invariant in $\widehat{\widehat{\mathcal{F}}}'$

so that if $g \geq 2$, we have canonically $\Gamma_{g,\nu} \simeq \Gamma_{g,0}$, on the other hand $\Gamma_{1,\nu} \simeq \Gamma_{1,1}$ $(g = 1, \nu \geq 1)$, $\Gamma_{0,\nu} \simeq \Gamma_{0,3}$ $(\nu \geq 3)$.

N.B. The "fact" † assumed is far from obvious — even that $\widehat{\widehat{\mathcal{F}}}^{!}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{F}}}_{g,\nu-1}|\widehat{\widehat{\mathcal{M}}}_{g,\nu-1}$ is surjective, or that $\widehat{\widehat{\mathcal{F}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1}$ is injective, is far from obvious, and is perhaps altogether false !

The assumed fact amounts to a statement of the existence of a functor in the opposite direction "drilling holes", which|what in any case remains not understood. [p. 348]

Uncertainty retained from cancelled passages.

140-2-s349-D001 giving by

140-2-s349-D002 $N(\Sigma)$

140-2-s349-D003 $\Gamma_{g,0}$

140-2-s349-D004 [illeg. 1]

[scan 350]

28. Change of type (g, ν) (continued) : passage to a finite covering. [p. 177]

Let π be a profinite group with loops of type (g, ν), anabelian as always, π' a subgroup of finite index. We know (or we|we thereby verify, by passing to the discrete case) that † equipped with the traces on|under π of the loop subgroups of π, π' is a group with loops.

†

The loop subgroups are also|thus the † subgroups L' such that

$$\text{a)} \quad L' = \mathrm{Centr}_\pi(L') \cap \pi'.$$

$$\text{b)} \quad L = | : \mathrm{Centr}_\pi(L') \text{ is a loop subgroup in } \pi.$$

We put $d(L') = [L : L']$, and we thus have a function

$$d : I' = I(\pi') \longrightarrow \mathbb{N}^*,$$

and a map $I' \overset{\varphi}{\to} I$, such that $\forall i \in I$, we have

$$(1) \qquad \sum_{\substack{i' \in I' \\ \text{over } i}} d(i') = n \qquad (\text{where|here } n = [\pi : \pi'])$$

We will have the Hurwitz formula

$$(2) \qquad \dagger\ 2g' - 2 = n(2g-2) + \overbrace{\sum_{i' \in I'} (d(i') - 1)}^{n \,\mathrm{card}\, I - \mathrm{card}\, I'}$$

[p. 349]

Uncertainty retained from cancelled passages.

140-2-s350-D001 [illeg. 1] discretified

140-2-s350-D002 n int(g) } n counted|to count int(g) $d_\alpha|d_{i\alpha}$ $L'_{i\alpha}|N'_{i\alpha}$ $\cap\pi'$ for conjugacy

140-2-s350-D006 the

140-2-s350-D007 $2g|2\alpha$